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WOMEN SAFETY HIDDEN MALICIOUS CHIP USING DEEP LEARNING AND IOT BASED TRACKING TECHNOLOGY

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ABSTRACT

Our Invention women safety hidden malicious chip using deep learning and IOT based tracking technology is a personal protective device is integrated with a human body and 4-G Mobile phone. The invention is also a activation of either device causes activation of the other device and to a activation is achieved by a switch or voice command or activation a message asking for assistance at the present location is sent to a previously agreed upon emergency contact. The Invented technology also includes a smartphone also starts both audio and video recording, which are also part of the data sent to the emergency contact person or service. The invention is also works protective devices are integrated into a unit that may combine such items as a protective hidden spray, high density Taser, hidden strobe light, and other deterrent technologies to ward off would-be attackers. The invention is to a multiple-bit microprocessor is configured and coupled for controlling functions of the wearable accessory the microprocessor housed within the housing and a storage means for storing data representative of the signals the storage means coupled with the microprocessor.

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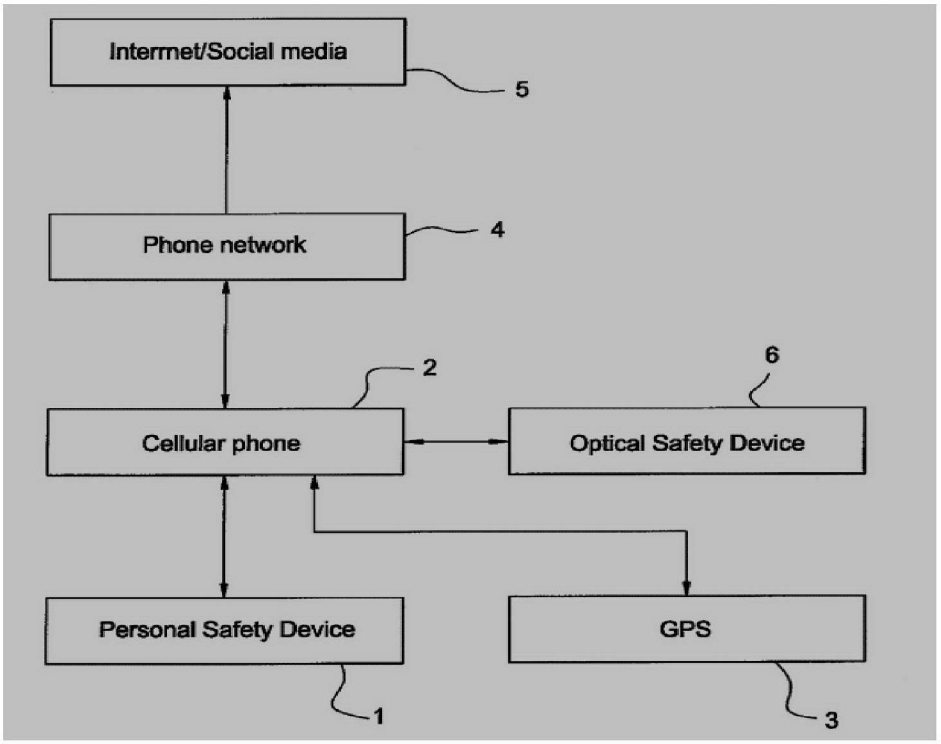




FIG. 1 is a schematic of the main components associated with the Personal Safety System (PSS).

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WOMEN SAFETY HIDDEN MALICIOUS CHIP USING DEEP LEARNING AND IOT BASED TRACKING TECHNOLOGY

FIELD OF THE INVENTION

Our Invention is related to a women safety hidden malicious chip using deep learning and IoT based tracking technology.

BACKGROUND OF THE INVENTION

The present invention relates to a wireless personal safety device. Every day, people are exposed to situations in which personal safety may be in jeopardy: while strolling on city sidewalks, hurrying to one's car at night, looking for one's car in a parking garage, jogging on a lonely road, walking the dog out at night, getting lost outdoors, taking a short cut down a deserted street, getting waylaid by a vicious dog, hearing footsteps following one, waiting for help when one's car breaks down, making a wrong turn in a dangerous neighborhood etc. In today's world people need a security companion.

Statistically 33% of women in the USA will be violently victimized at least once in their lifetimes. The Justice Department counted 4.9 million violent crimes involving over 2.7 million victims in 2010. It estimates there were over 467,000 firearm victims in 2011. Kids Fighting Chance, a children's advocacy group, reports one child is abducted or reported missing every 40 minutes or 800,000 times per year.

According to criminal deterrence theory, an attack occurs after the criminal makes a cost-benefit analysis. He will act if the benefit outweighs expected consequence. Criminals also know the chances that the police will be in earshot during an attack are low.

Calls to the police are not sufficient for deterrence. Most callers assume that law enforcement will respond before a criminal act occurs. Statistics indicate otherwise. Law enforcement rarely breaks up a street crime. More often than not, its role is pursuit, investigation and witness for prosecution of the criminal act.

The present invention is in the field of personal safety and more particularly for personal protective devices, particularly as combined with cellular telephones and more particularly when a smartphone is connected to a separate personal protective device via a network.

Historically the primary responsibility for personal protection has always fallen on the individuals. This is true regardless of the community where one lives. Police agencies have been unable to protect everyone. Today, many innocent people find themselves victims of crime. The nature of the crime varies from minor misdemeanors, such as theft, to violent assault, such as rape or murder. Typically, the victim of a crime is physically unable to prevent or deter an attack.

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Application of the personal attack alarm is aimed at persons at risk of being attacked outside their home for instance when individuals for one reason or another are out on the streets at night or live in insecure neighborhoods are suddenly attacked. These people may be the elderly, teenagers or women at risk of being attacked. Living in and around our large metropolitan cities requires that individuals, whether alone or with others, take precautionary measures to protect themselves against personal assaults. Various personal safety devices are available, including firearms, stun guns, chemical sprays, audible alarms and wireless security services.

Personal protection devices such as chemical sprays, for example, pepper sprays, audible personal security alarms, and stun guns, when used properly, all have a deterrent effect on an aggressor without the consequences associated with a firearm. These devices, however, have limitations and can prove to be ineffective depending upon the circumstances. Chemical sprays and audible alarms have their advantages and disadvantages.

Chemical sprays generally offer a user the advantage of deterring an assailant at a distance of as much as 10 to 15 feet, as well as giving notice to the assailant that the user is not totally defenseless. However, a chemical spray is of no use when the assailant covers his eyes with eyeglasses. Audible alarms on the other hand when used as a standalone device have lost their usefulness, since most people in the metropolitan city areas no longer pay attention to such sound generated devices. Moreover, as such, many people have resorted to carrying lethal weapons, such as hand guns. However, in certain jurisdictions, a person is forbidden by law to possess or use a lethal weapon.

Mobile and wireless technology have touched our lives in a multitude of ways ranging from communicating with near and dear ones to managing different aspects of our personal and business functions. Mobile technology is now also making a foray into personal safety management given the broad range of features and functions it possesses to facilitate things like capturing evidence at the scene of a crime in terms of both sound or visual evidence. It can also help with transmitting various signals for assistance.

The drawback with handheld devices is that users need to visually look at the phone in order to use it. It is very hard to try to use the mobile device in a concealed or hidden way. If the mobile device is in a person's pocket, bag or purse, it makes it that much more difficult to try to access it, look at the device and use it for a particular purpose. And one cannot figure out the current state of the device or operate any of the functions without actually looking at it.

OBJECTIVES OF THE INVENTION

1. The objective of the invention is to a personal protective device is integrated with a human body and 4-G Mobile phone?
2. The other objective of the invention is to a activation of either device causes activation of the other device and to a activation is achieved by a switch or

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voice command or activation a message asking for assistance at the present location is sent to a previously agreed upon emergency contact.

3. The other objective of the invention is to a smartphone also starts both audio and video recording, which are also part of the data sent to the emergency contact person or service.
4. The other objective of the invention is to a works protective devices are integrated into a unit that may combine such items as a protective hidden spray, high density Taser, hidden strobe light, and other deterrent technologies to ward off would-be attackers.
5. The other objective of the invention is to a multiple-bit microprocessor is configured and coupled for controlling functions of the wearable accessory the microprocessor housed within the housing and a storage means for storing data representative of the signals the storage means coupled with the microprocessor.

SUMMARY OF THE INVENTION

The present invention relates to a wireless personal safety device. In these respects, the combination cellular phone and personal safety device according to the present invention substantially departs from the conventional concepts and designs of the prior art, and in so doing provides an apparatus primarily developed for the purpose of providing a means of communication and a means of self-defense in one unit.

The present invention includes the following interrelated objects, aspects and features:

(1) In a first aspect, the present invention contemplates a wireless personal safety device that is self-contained but, when activated, communicates with a cellular telephone while also providing self-defense capability.

(2) In the preferred embodiment, the personal safety device includes a shocking mechanism and a sprayer designed to spray a substance for self-defense purposes such as pepper spray or tear gas.

(3) In order to operate the present invention, the personal safety software application is downloaded into a user's cellular telephone (cell phone). Among other things, in downloading the software, a variety of options may be selected such as, for example, contact information for the user, social media contact information, whether 911 emergency calling is to be enabled.

(4) In another aspect, the software provides the capability of activating a separation alarm. Thus, if the cell phone and the personal safety device are separated by a preset distance, for a pre-set time, an alarm can be triggered allowing a third party site such as the police to track the user's location.

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(5) The personal safety device may also be provided with a panic button which when depressed notifies the cell phone via Bluetooth.

(6) The system may include means for monitoring various capabilities such as, for example, checking of the level of the spraying substance as well as the level of charge of the personal safety device battery which impacts the amount of voltage that may be applied by the shocking mechanism. Additionally, the user can monitor the strength of signal between the personal safety device and the associated cell phone to make sure that signals sent by the personal safety device may be effectively received by the cell phone.

(7) In another aspect the present invention contemplates a wireless optical safety device that is self-contained but, when activated, communicates with a cellular telephone while also providing assailant detection capabilities.

(8) Through use of the present invention, a user may feel safe while carrying a personal safety device that can not only provide self-help through shocking or spraying an assailant, but may also communicate with a cell phone that permits authorities to quickly locate a crime victim and protect them from further harm.

As such, it is a first object of the present invention to provide a wireless personal safety device.

It is a further object of the present invention to provide such a device which wirelessly communicates with a cell phone to enable transmission of information concerning an attack to authorities and social media.

It is a further object of the present invention to provide such a system in which a personal safety device may be equipped with a shocking mechanism.

It is a still further object of the present invention to provide a personal safety device that may be equipped with a spraying apparatus designed to spray a noxious spray onto an assailant.

It is a further object of the present invention to provide such a device which may include a panic switch allowing manual activation of notification.

It is a yet further object of the present invention to provide such a device in which a sensor can sense the spacing between a personal safety device and an associated cell phone and trigger remote notification when the spacing exceeds a pre-set distance.

It is a still further object of the present invention to provide a wireless optical safety device.

It is a yet further object of the present invention to provide a wireless optical safety device that is self-contained but, when activated, communicates with a cellular telephone while also providing assailant detection capabilities.

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The inventive system and method enhances the process of notifying an emergency situation by allowing a user to send audio (voice based), video, images (pictures) and text based data in addition to the voice based communication by triggering a distress signal using a wearable accessory. Accordingly, one aspect of the present invention relates to a method which enables a user to customize and notify multiple pre-configured members in the user's defined network.

In one embodiment, the method follows the steps of (a) triggering an emergency assistance signal by the user via at least one wearable accessory that includes a circuit enclosed by a casing. The circuitry (enclosed in the casing) is removable and can be attached to any other wearable accessory, (b) activating at least one handheld device, upon receipt of the triggered emergency assistance signal from the wearable accessory and establishing a communication thereof using a communication protocol, (c) recording all the events including audio and video at the wearable accessory, transmitting this to a handheld device which in turn transmits the same in real-time to a remote server, (d) receiving an acknowledgement signal from the handheld device to the wearable accessory using the communication protocol, wherein the acknowledgement signal includes a vibration to the wearable accessory which provides confirmation to the user that the emergency assistance signal has been triggered successfully and the same has been sent to a list of pre-configured members stored in the handheld device, and (e) sending alert messages with Global Positioning System (GPS)/Indoor Positioning System (IPS) location to the list of pre-configured emergency contacts of the user via social media, email, text messages, voice message, voice call etc.

Another aspect of the present invention provides a wearable accessory, including housing, a vibration generation means for producing a vibration. The vibration generation means housed within the housing has at least one switch for allowing a user of the wearable accessory to cause activation of the vibration generation means.

The switch is housed within the housing and there is a module for communicating with at least one handheld device using a communication protocol, an image capturing device including a microphone for receiving audio and video signals at the wearable accessory, a USB port for charging the wearable accessory, a multiple-bit microprocessor configured and coupled for controlling functions of the wearable accessory. The microprocessor is housed within the housing and there is a storage mechanism for storing data representative of the signals, the storage mechanism is coupled with the microprocessor.

BRIEF DESCRIPTION OF THE DIAGRAM

FIG. 1 is a schematic of the main components associated with the Personal Safety System (PSS).

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FIG. 2 shows a flow chart of a method which enables a user to notify multiple pre-configured members in the user's defined network and receive responses back from the pre-defined network, according to one embodiment of the present invention.

FIG. 3 is a schematic of the Optical Safety Device of the present invention.

FIGS. 4A and 4B show a flowchart of the application installer software on the cellular phone.

DESCRIPTION OF THE INVENTION

FIG. 1 represents the main components of the Personal Safety System (PSS) invention. The personal safety device (PSD) 1 is battery powered (107) and contains self-defense mechanisms such as a shocking mechanism and a spray mechanism to subdue or discourage a would-be attacker. The PSD 1 is tethered to a cellular phone 2, over a Bluetooth system. The optical safety device (OSD) 6 is battery powered (287) (FIG. 3) and contains camera component used to identify and track potential assailants coming from behind. The OSD 6 is tethered to a cellular phone 2, over a Bluetooth system 290, 291 (FIG. 3).

The cellular phone 2 contains an internal Global Positioning System (GPS) 3 that provides location information of the whereabouts of the cellular phone, and therefore presumably of the cellular phone owner. The cellular phone 2 communicates over a cellular phone network 4 to provide interconnectedness to the phone network and the Internet 5. The phone network 4 allows the user to summon help over the phone network (calling 911 or calling another person directly). The Internet 5 also has linkages to various social media for summoning help over social media, such as, for example, Facebook. Also, as is known in the art, GPS is sometimes aided by location information obtained by triangulation involving cellular phone towers. U.S. Pat. No. 5,515,419 discloses this aspect.

FIG. 2. At step 201, an emergency assistance signal is triggered from the wearable accessory having an electronic circuit embedded therein. At step 202, the triggered signal is transmitted to a handheld device thereby activating the device to establish a communication between the device and the wearable accessory using any communication protocol, the communication protocol may include but is not limited to Bluetooth, Bluetooth Low Energy or RFID etc. At step 203, the handheld device automatically starts the application which is a pre-installed application once it receives a command from the circuit through the communication protocol.

At step 204, the circuit continuously checks, whether the user has cancelled the emergency assistance signal in order to clarify whether the emergency signal has been wrongly initiated or not. If yes, at step 205, the circuit waits for a predetermined time interval and cancels the activation of the emergency assistance signal, and the wearable accessory and the handheld device return backs to its

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normal mode. The wearable accessory records all the events including audio or a video or both with the help of the electronic circuit therein and receives a vibration confirming cancellation. At step 206, the triggered signal received at the handheld device generates a distress signal by way of sending alert messages and GPS/IPS location to pre-configured emergency contacts of the user by social media, email, text messages and pre-configured voice messages.

In an embodiment of the present invention, the location of the user is determined by Indoor Positioning System (IPS) which is based on magnetic sensor data from the handheld device which is used to wirelessly locate the user inside a building. By using the IPS, one can find the exact location of the user within the building, where the GPS fails to establish the exact indoor locations. Further, the circuitry of the wearable accessory communicates to a remote server with the help of the handheld device or communication device. Upon receiving at the server, in response, at step 207, an acknowledgement signal is received at the wearable accessory via the handheld device.

The acknowledgement signal includes a vibration to the wearable accessory in order to provide a confirmation to the user that the emergency assistance signal has been triggered successfully and the same has been sent to the list of pre-configured emergency contacts on the handheld device. The distress signal is transmitted by the handheld device to other location using a network for e.g. a victim's network. It is also possible that the distress signal can be sent simultaneously to all other available networks which are in sync with the electronic circuit of the wearable accessory.

In Step 208 of the present invention, at least one person from the user's personal network responds to the emergency notification they received via Step 206. The response might be in the form of text, SMS, MMS, Social Media alert such as facebook, etc. An acknowledgement signal is then received at the wearable accessory via the handheld device. The acknowledgement signal includes a vibration to the wearable accessory in order to provide a confirmation to the user that help is on the way. In another aspect of the present invention, the user's personal network is sent a message periodically with user's updated location where law enforcement can track user on a web-based application, as step 209.

Once help reaches user, the user can cancel the alert via wearable accessory or via the handheld device, as step 210. Upon doing that, the user receives cancellation vibration on wearable accessory and also user's network is informed that the user has cancelled emergency, as in step 211. Further, the emergency log with full event information is created and stored on server and handheld device, as in step 211.

FIG. 3, the camera 295 is an infrared (IR) camera, useful for viewing in darkness and IR light. IR light is provided by IR LEDs 294. This light is invisible to humans, thereby also concealing the fact that the OSD 281 is active. While other cameras can be used, the IR camera provides less data to process (monochromatic vs. color), simplifying the detection of a rear approaching assailant. Ambient light

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sensor 293 is a semiconductor device that senses the ambient light level. This information is incorporated in the camera interface/logic module 292 to control the intensity of the IR LEDs 294. When daylight is sensed the IR LEDs 294 are turned off. As the ambient light decreases the IR LEDs 294 are slowly turned up in intensity so that the IR camera 295 always has adequate light for assailant detection. Image data from the camera 295, suitably processed by interface 292, is sent to Bluetooth module 291, where it is transmitted to the cellular phone 2 for image processing and analysis. As shown by the arrows, information from the cellular phone 2 can also provide some control over the camera 295 such as update rate, aperture, etc. Referring now to the invention in more detail, in FIGS. 4A and 4B there is shown a flowchart of the software installer used to install the PSS application onto the cellular phone. Block 310 represents the user installing the application on the cellular phone. As is known in the art, this is accomplished by downloading the application from one of the various vendors' software repositories, such as the Apple store or Google Play. Once the application is installed, the user runs the application in Block 320. The user is then prompted to set various features and options that control how the cellular phone connects with other users of the network and how the cellular phone interacts with the PSD, as denoted by Block 330.

Various options are now selected. In Block 340, the user is asked if they would like to enter/add contact information. Such information can be a phone number, cellular and/or landline, an address, either physical and/or e-mail, etc. If the answer is yes, the user enters the desired data in Block 345. In Block 350, the user is next asked if they would like to enter social media contact information. Such information can be links for posting to Facebook, Twitter, and other social media sharing sites. If the answer is yes, the user enters the desired data in Block 355. In Block 360, the user is asked if they would like to add 911 emergency calling. If the answer is yes, the information is saved in Block 365.

In Block 370, the user is asked whether to enable the separation sensor and alarm. This alarm is triggered when the PSD and the cellular phone are separated by a certain distance. This distance is based on the received signal strength indicator (RSSI) indicator. There is no standardized relationship between range and the RSSI reading, so this can only be used as an approximate distance measure, but it is sufficient. If enabled, the approximate distance is entered in Block 375. Additionally, related timing information can be entered, such as how long the devices are separated before enabling the alarm, giving the owner time to bring the PSD and cellular phone together again. In Block 380, the user is asked what capture methods should be enabled to capture audio, video and location information.

If selected, the user enters or checks the various methods. Block 390 asks the user whether location sharing should be enabled. Location sharing is a feature that allows you to let people know where you are. Both Wi-Fi positioning system and GPS are used to pinpoint the user's location as accurately as possible. If it is desired, then it is enabled in Block 395. Block 400 is then reached where Bluetooth is enabled in the cellular phone. Once enabled, the user is asked to turn on the PSD

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and/or the OSD so that the cellular phone can be paired to the appropriate devices in Block 410. This is a standard method of using Bluetooth as is known in the art.

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WE CLAIM

1) Our Invention women safety hidden malicious chip using deep learning and IOT based tracking technology is a personal protective device is integrated with a human body and 4-G Mobile phone. The invention is also a activation of either device causes activation of the other device and to a activation is achieved by a switch or voice command or activation a message asking for assistance at the present location is sent to a previously agreed upon emergency contact. The Invented technology also includes a smartphone also starts both audio and video recording, which are also part of the data sent to the emergency contact person or service. The invention is also works protective devices are integrated into a unit that may combine such items as a protective hidden spray, high density Taser, hidden strobe light, and other deterrent technologies to ward off would-be attackers. The invention is to a multiple-bit microprocessor is configured and coupled for controlling functions of the wearable accessory the microprocessor housed within the housing and a storage means for storing data representative of the signals the storage means coupled with the microprocessor.
2) According to claim1# the invention is to a personal protective device is integrated with a human body and 4-G Mobile phone?
3) According to claim1,2# the invention is to a activation of either device causes activation of the other device and to a activation is achieved by a switch or voice command or activation a message asking for assistance at the present location is sent to a previously agreed upon emergency contact.
4) According to claim1,2,3# the invention is to a smartphone also starts both audio and video recording, which are also part of the data sent to the emergency contact person or service.
5) According to claim1,2,3,4# the invention is to a works protective devices are integrated into a unit that may combine such items as a protective hidden spray, high density Taser, hidden strobe light, and other deterrent technologies to ward off would-be attackers.
6) According to claim1,2,4,5# the invention is to a multiple-bit microprocessor is configured and coupled for controlling functions of the wearable accessory the microprocessor housed within the housing and a storage means for storing data representative of the signals the storage means coupled with the microprocessor.

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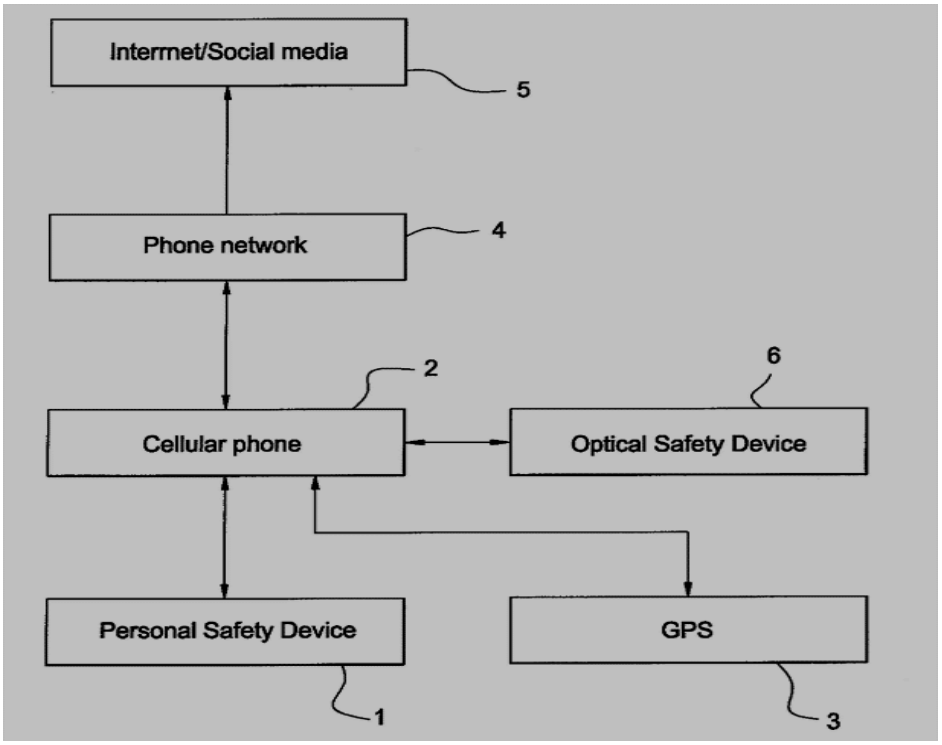




FIG. 1 is a schematic of the main components associated with the Personal Safety System (PSS).

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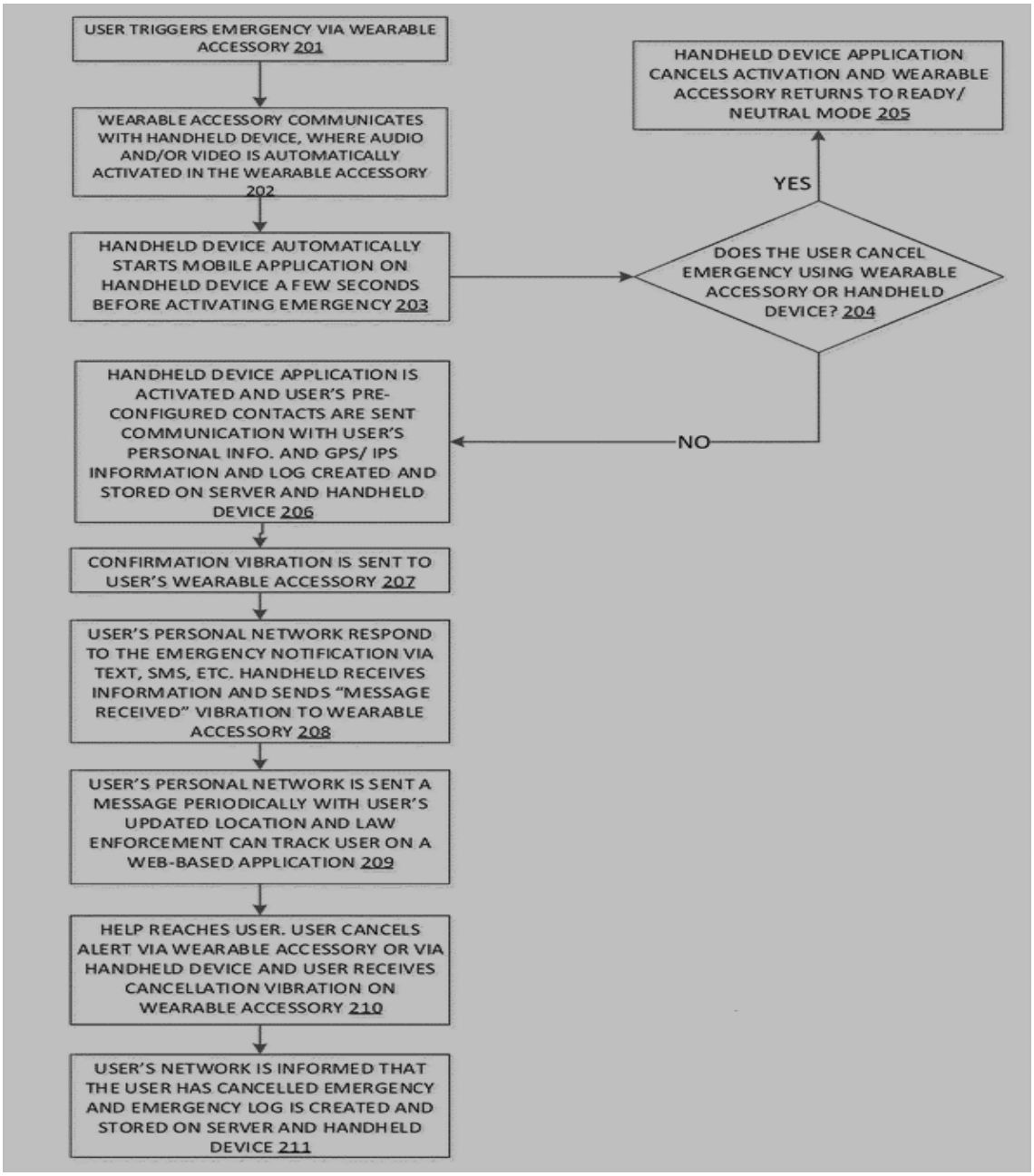


FIG. 2 shows a flow chart of a method which enables a user to notify multiple pre-configured members in the user's defined network and receive responses back from the pre-defined network, according to one embodiment of the present invention.

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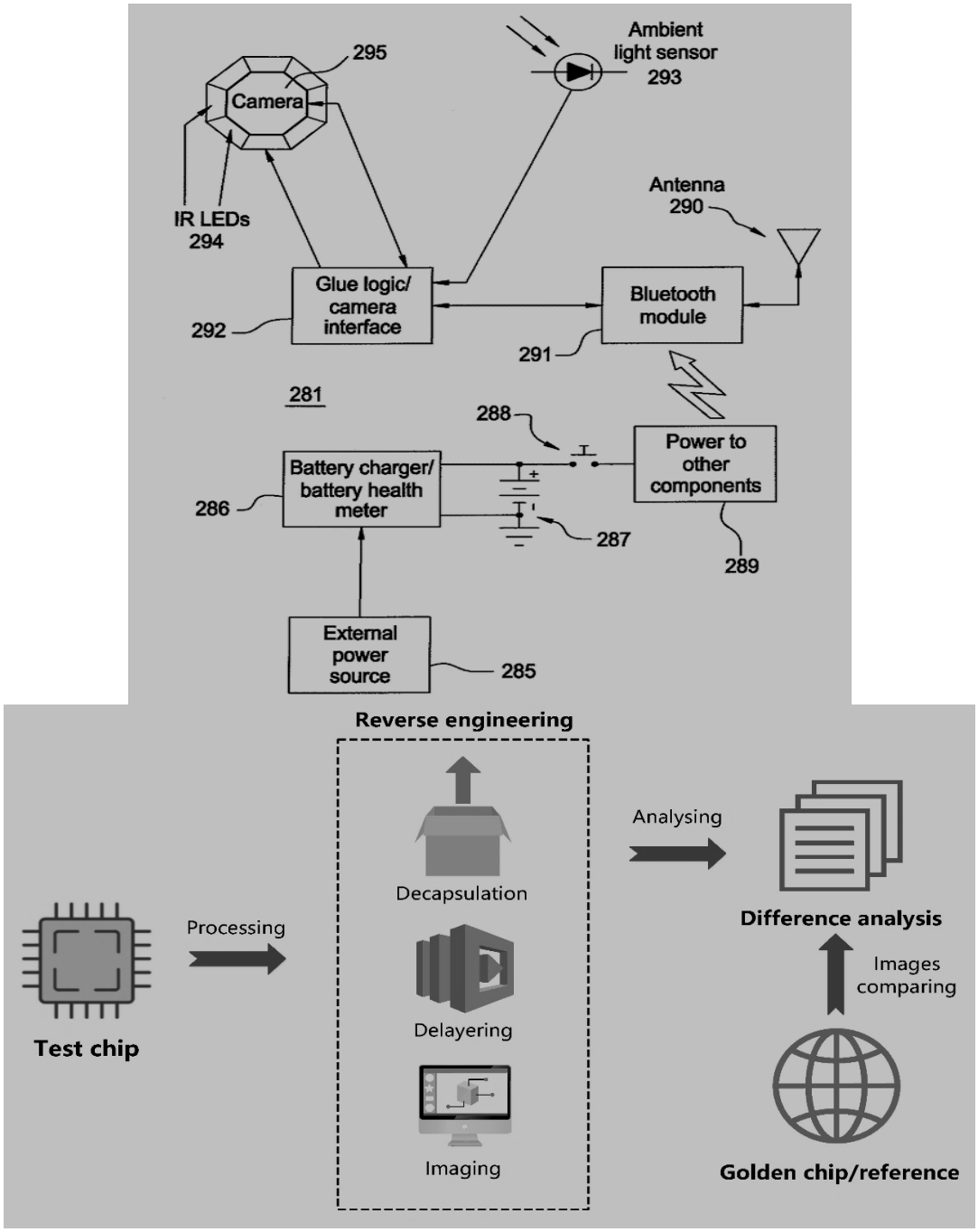


FIG. 3 is a schematic of the Optical Safety Device of the present invention.

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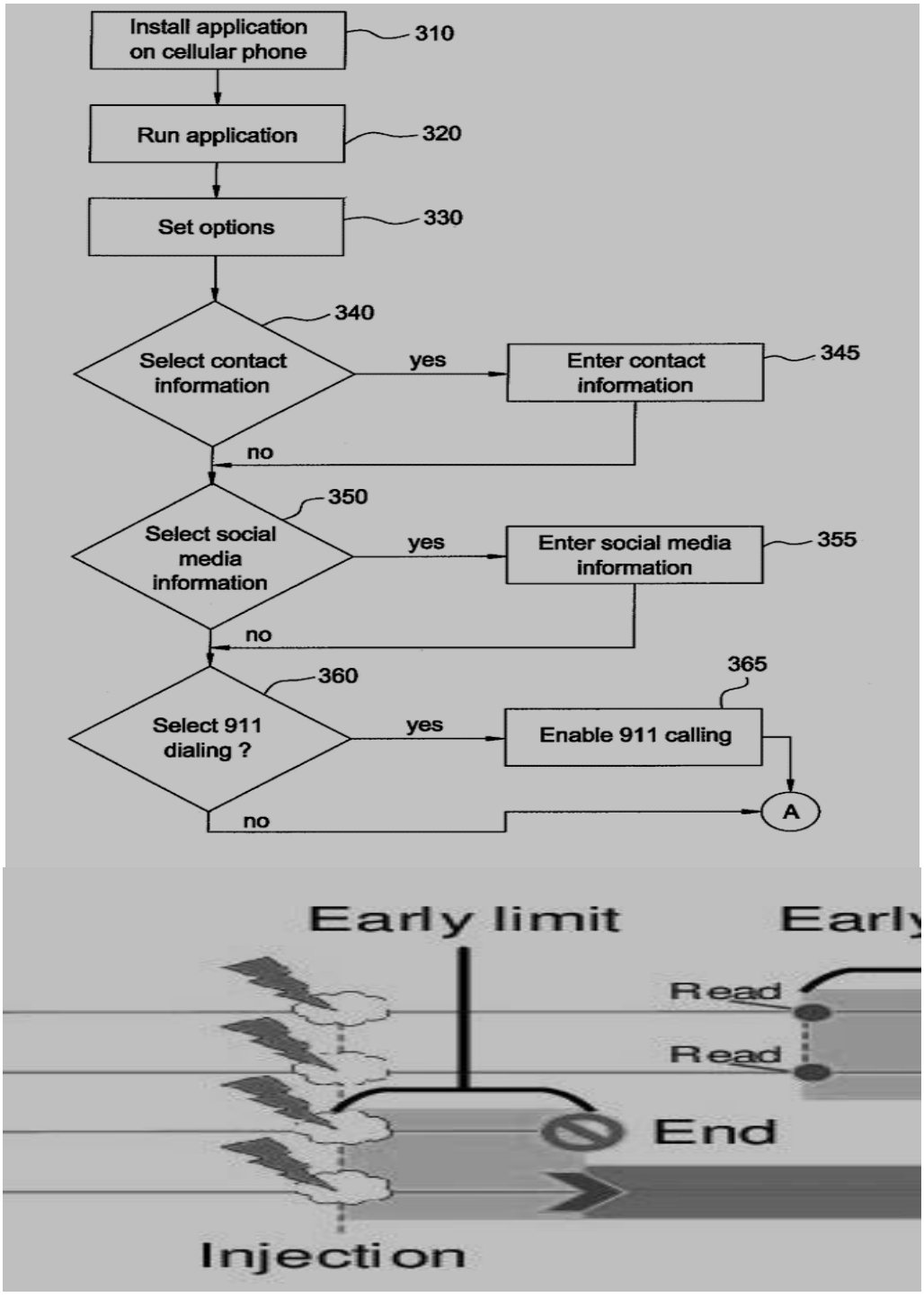


FIG. 4A: show a flowchart of the application installer software on the cellular phone.

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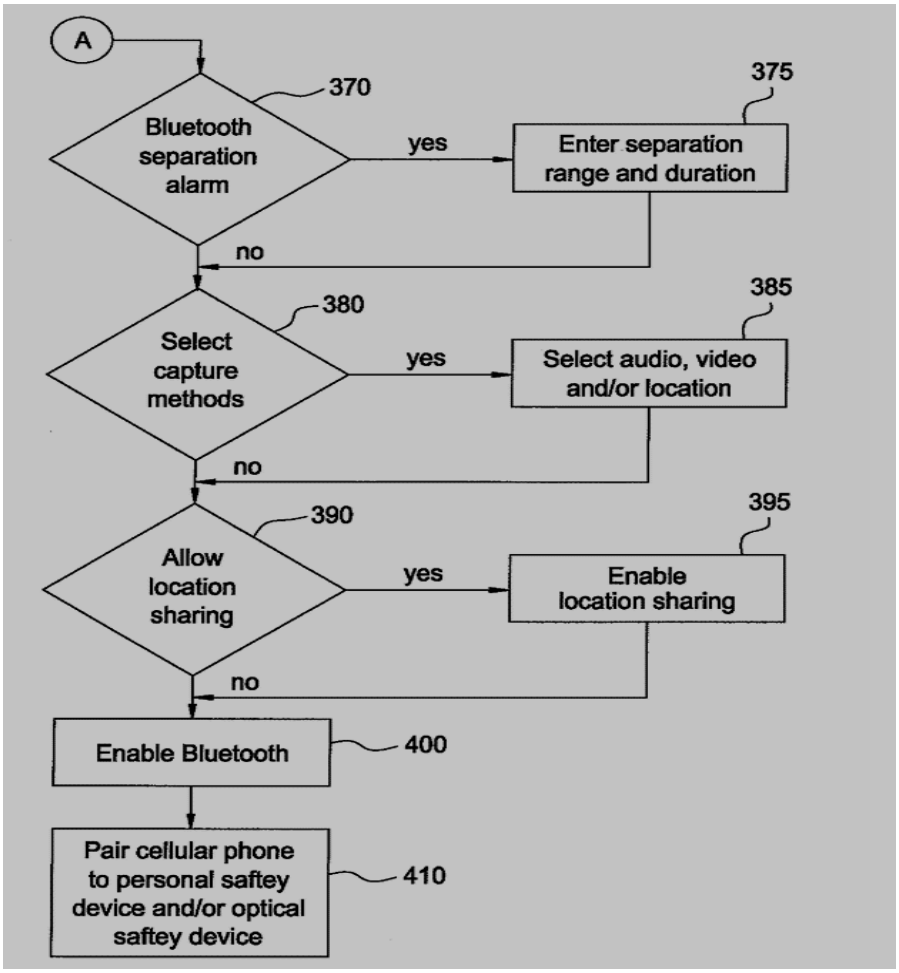


FIG. 4B show a flowchart of the application installer software on the cellular phone.